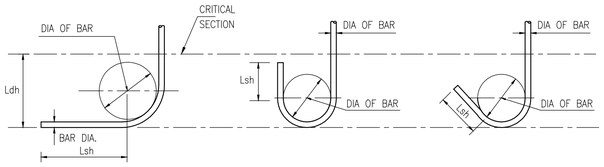
DRAWING NUMBER	REV
MD-216-1000-EG-CV-DED-1005	D01

1.0 DEVELOPMENT LENGTH OF STANDARD HOOKS

1.1 TABLE OF DEVELOPMENT LENGTH OF STANDARD HOOKS

BAR DIA (MM)	fy (MPa)	f'c (MPa)	Ldh (MM)	Lsh (FOR MAIN BAR) (MM)		Lsh (FOR STIRRUP/TIE) (MM)	
				90° HOOK	180° HOOK	90° HOOK	135° HOOK
D10	420	35	150	150	100	100	100
D12	420	35	150	200	150	150	150
D16	420	35	200	200	150	150	150
D20	420	35	250	250	150	-	-
D22	420	35	250	300	200	-	-
D25	420	35	300	300	200	-	-
D28	420	35	300	350	250	-	-
D32	420	35	350	400	300	-	-

LEGEND : Ldh = DEVELOPMENT LENGTH IN TENSION OF HOOKED BAR
Lsh = STANDARD HOOK LENGTH OF DEFORMED BAR



2.0 LAP SPLICE & DEVELOPMENT LENGTH

UNLESS NOTED OTHERWISE ON DRAWINGS, ALL REINFORCEMENT STEEL SPLICES & ANCHORAGE LENGTHS SHALL BE IN ACCORDANCE WITH ACI 318-14M AS PER FOLLOWING TABLE.

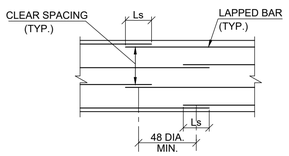
2.1 TABLE OF DEVELOPMENT LENGTH (Ld)

BAR DIA (MM)	fy (MPa)	f'c (MPa)	Ld (MM)	
			TOP	OTHER
D10	420	35	230	200
D12	420	35	280	240
D16	420	35	370	320
D20	420	35	460	400
D22	420	35	900	700
D25	420	35	1030	800
D28	420	35	1150	900
D32	420	35	1320	1030

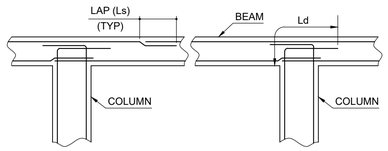
2.2 TABLE OF LAP SPLICE LENGTH (Ls)

BAR DIA (MM)	fy (MPa)	f'c (MPa)	Ls (MM)	
			TOP	OTHER
D10	420	35	400	400
D12	420	35	480	480
D16	420	35	640	640
D20	420	35	800	800
D22	420	35	1210	880
D25	420	35	1380	1000
D28	420	35	1550	1120
D32	420	35	1750	1280

NOTES :
SPLICES IN REINFORCEMENT SHALL BE PROVIDED IN THOSE POSITIONS SHOWN IN THE DESIGN DRAWINGS. IN ALL CONCRETE ELEMENTS WITH LONG CONTINUOUS BARS, SPLICES SHALL BE STAGGERED IN ADJACENT BARS (START TO START) BY 48xBAR DIAMETER MINIMUM. SPLICES SHALL NOT BE PROVIDED WHERE CRITICAL BENDING STRESSES ARE DEVELOPED.

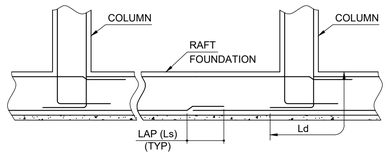


2.3 LAP SPLICE IN BEAMS



PREFERRED LOCATION OF LAP SPLICE IN BEAMS
(UNLESS NOTED OTHERWISE IN DESIGN DRGS.)

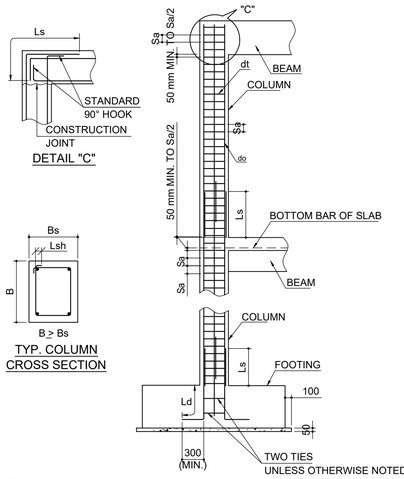
2.4 LAP SPLICE IN FOUNDATIONS



PREFERRED LOCATION OF LAP SPLICE IN RAFT FOUNDATION
(UNLESS NOTED OTHERWISE IN DESIGN DRGS.)

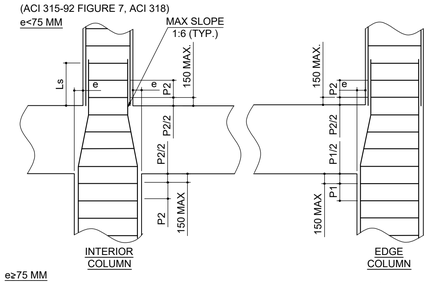
3. STANDARD BAR ARRANGEMENT

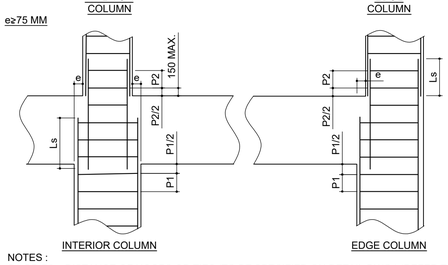
3.1.1 COLUMN AND FOOTING
STANDARD BAR ARRANGEMENT (ACI 315-92 FIGURE 7, ACI 318)



NOTES :
Ls = LAP SPLICE LENGTH FOR COMPRESSION BAR (REFER TO 2.2)
Ls = DEVELOPMENT LENGTH OF STANDARD HOOK IN TENSION (REFER TO 1.1)
Sa = DISTANCE OF HOOPS OR TIES (TO BE SPECIFIED IN DETAIL DWG. CONFORMING TO ACI-318)
Sa ≤ 16do, 48dt & LEAST DIMENSION OF COLUMN (ACI-318)

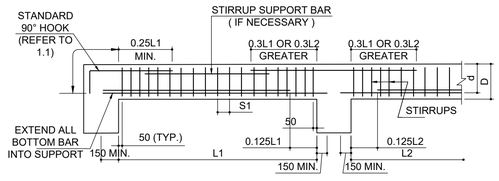
3.1.2 COLUMN OFFSET
(ACI 315-92 FIGURE 7, ACI 318)



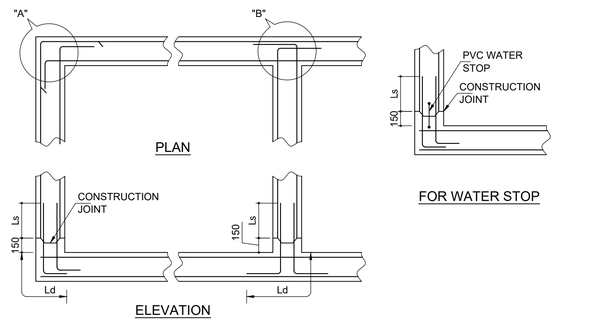


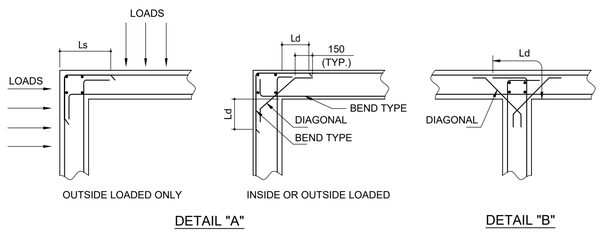
NOTES :
P1 OR P2 = DISTANCE OF HOOPS OR TIES (TO BE SPECIFIED ON DETAIL DWG.) (REFER TO 2.2)
Ls = LAP SPLICE LENGTH

3.2 BEAMS AND GIRDERS
STANDARD BAR ARRANGEMENT FOR BEAMS (ACI 315-92 FIGURE 2)



3.3 STANDARD BAR ARRANGEMENT FOR WALL

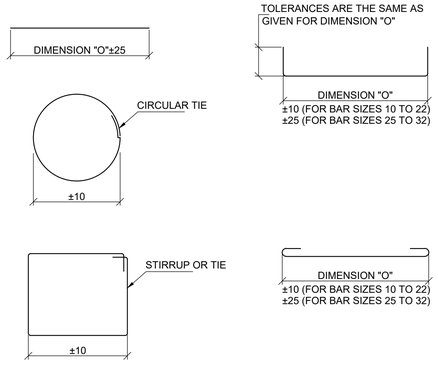




4.0 BAR BENDING DETAIL

1) FABRICATION TOLERANCES & STANDARD HOOK DETAILS ARE GIVEN IN FIGURE 1.
2) ALL HOOKS ARE IN ACCORDANCE WITH THE REQUIREMENTS OF ACI 318 "BUILDING CODE REQUIREMENTS FOR REINFORCED CONCRETE".

FIGURE 1 : FABRICATION TOLERANCES
ALL DIMENSIONS ARE MEASURED FROM OUTSIDE TO OUTSIDE OF BARS.



NOTES :-

1. ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE STATED.
2. DIMENSION "O" IS MEASURED FROM OUTSIDE TO OUTSIDE OF BARS.
3. THE RELEVANT PROVISIONS OF ACI 318-14 & SBC 304-18 HAS BEEN COMPLIED.

REV	DESCRIPTION	DRN	CHK	ENG	APP	EMP
D01	AS-BUILT	ABS	SJM	AVG	AG	
C01	ISSUED FOR CONSTRUCTION & REVISED AS PER	ABS	SJM	AVG	AG	
	EMPLOYER COMMENTS ON 20/04/2020					
A01	ISSUED FOR REFERENCE	ABS	SJM	AVG	AG	

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE
MD-216-1000-EG-ST-CRT-1001	CIVIL, STRUCTURAL & ARCHITECTURAL - DESIGN CRITERIA		
MD-216-1000-EG-CV-DED-1001	GENERAL NOTES FOR CIVIL WORKS	A01	24/03/2020

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OUTOTEC DRAWING NUMBER	900707__GAU01__ZZ01__CTB17__10010	DOCUMENT ID OU100676690 / REVISION 2
DRAWN	ABS	19/03/2020
CHECKED	SJM	21/03/2020
ENGINEER	AVG	20/03/2020
APPROVED	AG	24/03/2020
EMPLOYER		
SCALE	N.T.S.	A1

MANSOURAH - MASSARAH GOLD PROJECT
DETAIL DRAWING
GENERAL NOTES
STANDARD REINFORCEMENT DETAILS

DRAWING NUMBER: MD-216-1000-EG-CV-DED-1005 REV D01